Corrective & Preventive Action

AutoSaarthi | generated 08 Sep 2026 05:29 | run capa-2e14c0

CAPA ID	CAPA-DRAFT-20260908-2024-Q3-WP
Component	water_pump
Defect mode	water pump failures in BATCH-2024-Q3-WP at 10.3x the fleet baseline
Batch	BATCH-2024-Q3-WP
Plant	Nashik
Status	draft
Affected vehicles	7

Root cause

7 of 46 units failed (15.2% vs 1.5% baseline, p=6e-05, q=0.00249). Confidence: high.

Corrective action

Rework the affected units at the regional service network.

Preventive action

Add an end-of-line flow test at the sub-assembly station.

Five whys

1. {'why': 'Why are water pump failures elevated in BATCH-2024-Q3-WP?', 'because': '7 of 46 vehicles in that lot (15.2%) have had the water pump replaced, against a fleet baseline of 1.5%.', 'evidence': 'lift 10.3x, p=6e-05, q=0.00249'}

2. {'why': 'Why is this lot different from the rest of the parc?', 'because': 'The failures are confined to units built at Nashik within this batch, which points at a build- or supply-side change rather than in-service duty.', 'evidence': 'cohort scoped to water_pump | Nashik | BATCH-2024-Q3-WP'}

3. {'why': 'Why would a build-side change produce this failure mode?', 'because': 'Water Pump - impeller seal weep', 'evidence': 'matches historical CAPA-2022-1096 (similarity 0.3068)'}

4. {'why': 'What was the mechanism in that precedent?', 'because': 'A revised seal specification was released to production before the validation build completed.', 'evidence': 'CAPA-2022-1096 root-cause section'}

5. {'why': 'Why was it not caught before shipping?', 'because': 'The FMEA did not rank this failure mode above the action threshold.', 'evidence': 'CAPA-2022-1096 5-Why, final step'}

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